Modeling the emergence of vaccine-resistant variants with Gaussian convolution

COVID-19: Could the wrong strategy ruin vaccine efficiency?

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Part I: Article Approach : Optimal Control

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Part II: A Reinforcement Learning Approach

Part I

Vaccination vs NPI under variant evolution

Introduction

Introduction

- Vaccination reduces infection by the original strain.
- But if infection remains high, vaccination may create a selection pressure toward more vaccine-resistant variants.
- The paper asks:

Question

Is “vaccinate as fast as possible” always optimal, or can it backfire if NPIs are lifted too early ?

KEY MODELING IDEA

- Variants are classified by two traits:
  - transmissibility,
  - resistance to vaccine.
- Instead of one infected compartment $I(t)$, the model uses a matrix

$$I(t) = \big(I_{ij}(t)\big)_{1 \le i \le N,\ 1 \le j \le M}.$$

- Each entry $I_{ij}(t)$ = fraction of population infected by variant (i, j).

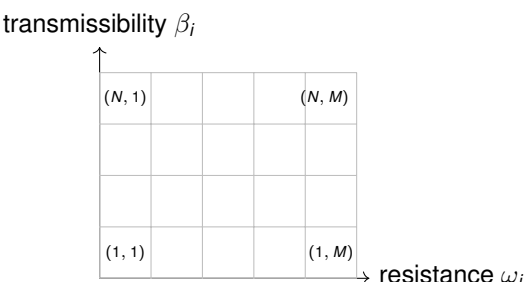
Model

VARIANT GRID

- Variants are classified by two traits:
  - transmissibility : β_i,
  - resistance to vaccine : ω_j.
- Instead of one infected compartment $I(t)$, the model uses a matrix

$$I(t) = \big(I_{ij}(t)\big)_{1 \le i \le N,\ 1 \le j \le M}.$$

- Each entry $I_{ij}(t)$ = fraction of population infected by variant (i, j).



$$\omega_j = \frac{j-1}{M-1}, \qquad \beta_i \text{ increasing with } i$$

Compartments

$$S(t), \quad V(t), \quad I(t) = \big(I_{ij}(t)\big), \quad R(t) = \big(R_{ij}(t)\big), \quad D(t)$$

- $S(t)$: susceptible
- $V(t)$: vaccinated
- $I_{ij}(t)$: infected by variant (i, j)
- $R_{ij}(t)$: recovered from variant (i, j)
- $D(t)$: deceased

Remark

The matrix structure is what lets the model represent both **infectiousness** and **vaccine resistance** at the same time.

Model Structure

- $n(t)$: vaccination rate
- $\eta(t) \in (0, 1]$: NPI coefficient
  - $\eta = 1$: no restriction
  - smaller η: stronger restrictions
- μ: fatality rate
- γ: recovery rate
- ξ_{ijkl}: reinfection tensor
- P: mutation tensor

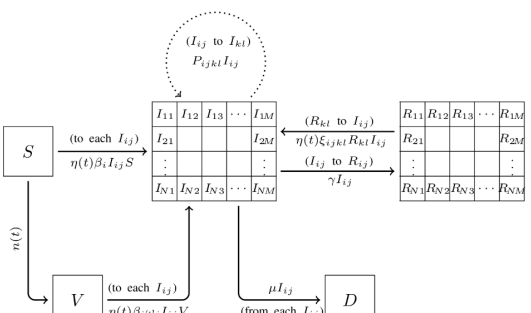


Figure 1. Model Diagram : representation of state transitions

Compact dynamical system

Using tensor notation, the model is

$$
\begin{aligned}
\frac{dS}{dt}(t) &= -n(t) - \eta(t)\big[(\beta \otimes 1) \cdot\cdot I(t)\big] S(t) \\
\frac{dV}{dt}(t) &= n(t) - \eta(t)\big[(\beta \otimes \omega) \cdot\cdot I(t)\big] V(t) \\
\frac{dI}{dt}(t) &= \eta(t)\big[(\beta \otimes 1) \circ I(t)\big] S(t) + \eta(t)\big[(\beta \otimes \omega) \circ I(t)\big] V(t) \\
&\qquad - \mu I(t) - \gamma I(t) + \eta(t)\big(\xi R(t)\big) \circ I(t) + \Psi\big(P \cdot\cdot I(t) - I(t)\big) \\
\frac{dR}{dt}(t) &= \gamma I(t) - \eta(t)\big(\xi^{(3,4,1,2)} I(t)\big) \circ R(t) \\
\frac{dD}{dt}(t) &= \mu I(t) \cdot\cdot (1 \otimes 1)
\end{aligned}
$$

Gaussian mutation operator

Schematically:

$$\frac{dI_{ij}}{dt} = \underbrace{\eta\beta_i I_{ij} S}_{S\to I_{ij}} + \underbrace{\eta\beta_i\omega_j I_{ij} V}_{V\to I_{ij}} - \underbrace{(\mu+\gamma) I_{ij}}_{\text{exit}} + \underbrace{\text{reinfection}}_{\xi} + \underbrace{\text{mutation}}_{P,\Psi}$$

Mutation is modeled as a Gaussian blur on the matrix I with a cutoff (to prevent the emergence of dangerous variant at the very beginning).

$$G_\sigma(t) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{t^2}{2\sigma^2}\right)$$

Optimal control

Optimal control problem

Controls:

$$u(t) = (n(t), \eta(t)).$$

Constraints:

$$\sup_{t\in[0,T]} n(t) \leq n_{\mathrm{SUP}}, \qquad \inf_{t\in[0,T]} \eta(t) \geq \eta_{\mathrm{INF}}.$$

Cost to minimize:

$$D(T) = \mu \int_0^T I(t) \cdot\cdot (1 \otimes 1)\, dt.$$

Interpretation

Choose the vaccination intensity and NPI level to minimize mortality.

Hamiltonian

State:

$$x(t) = \big(S(t), I(t), R(t), V(t)\big), \qquad p(t) = \big(p_S(t), p_I(t), p_R(t), p_V(t)\big).$$

Hamiltonian:

$$H(x, u, p) = \mu I \cdot\cdot (1 \otimes 1) + p_S \frac{\partial S}{\partial t} + p_I \cdot\cdot \frac{\partial I}{\partial t} + p_R \cdot\cdot \frac{\partial R}{\partial t} + p_V \frac{\partial V}{\partial t}.$$

Pontryagin condition:

$$\frac{\partial p}{\partial t}(t) = -\nabla_x H(x(t), u(t), p(t)), \qquad p(T) = 0.$$

Switching functions

Differentiating the Hamiltonian with respect to controls gives:

$$\Lambda_n(t) = \frac{\partial H}{\partial n} = p_V - p_S$$

and

$$\begin{aligned}\Lambda_\eta(t) = \frac{\partial H}{\partial \eta} = &-p_S\big((\beta \otimes 1) \cdot\cdot I(t)\big) S(t) - p_V\big((\beta \otimes \omega) \cdot\cdot I(t)\big) V(t) \\ &- p_R \cdot\cdot \big(\xi^{(3,4,1,2)} I(t) \circ R(t)\big) \\ &+ p_I \cdot\cdot \Big(((\beta \otimes 1) \circ I(t)) S + ((\beta \otimes \omega) \circ I(t)) V + \xi R(t) \circ I(t)\Big).\end{aligned}$$

BANG-BANG STRUCTURE AND CONTROL SQUARE

A necessary optimality condition is:

$$u^*(t) \in (\{0, n_{\mathrm{SUP}}\} \times [\eta_{\mathrm{INF}}, 1]) \cup ([0, n_{\mathrm{SUP}}] \times \{\eta_{\mathrm{INF}}, 1\}).$$

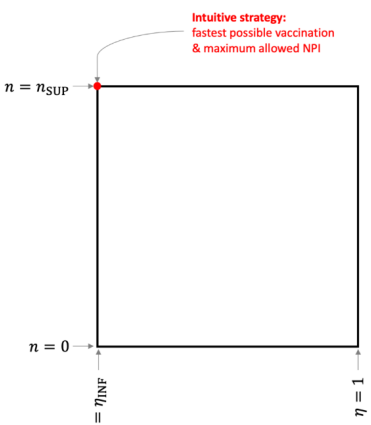


Figure 2. Bang-bang control : induced geometry of the solution

How the paper computes the optimum

Method 1: adjoint-based iteration

- start from the intuitive policy;
- compute the adjoint state;
- update the controls using the signs of Λ_n and Λ_η;
- smooth the updates with a relaxation parameter.

Method 2: simulated annealing

- discretize the control square;
- search for a near-optimal switching policy.

Results and interpretation

Simulation scenarios

Three scenarios are simulated:

1. low infection, no NPI;
2. high infection, no NPI;
3. high infection, with NPI initially.

Typical parameters:

$$N = 30, \qquad M = 40, \qquad R_0^{\max} = 15, \qquad \gamma = 0.09, \qquad \mu = 0.02.$$

$$n = 0.006 \quad \text{(0.6\% of the population vaccinated per day)}$$

Qualitative result

Vaccination works in low prevalence, and in high prevalence only if NPIs are maintained at least initially.

Main simulation insight

- High infection + vaccination + no NPI:
  - the original strain is reduced,
  - but resistant variants are selected,
  - a second peak appears.
- High infection + vaccination + initial NPI:
  - variants do not get enough time to emerge,
  - pandemic is controlled.

Counterintuitive result

- In the high-infection / no-NPI scenario, the intuitive strategy is not optimal.
- The near-optimal strategy delays vaccination to limit selection of highly resistant variants.
- This still has a large death toll, but lower than the naive fast-vaccination policy without NPIs.

Message

The short-term best action is not always the long-term best action when evolution is present.

TAKEAWAYS

- The core novelty is replacing the scalar infected compartment by the matrix I_{ij}.
- Mutation is modeled by Gaussian convolution in variant space.
- The optimal control problem is solved with Pontryagin + numerical search.
- Vaccination and NPIs must be analyzed jointly because they shape **selection pressure**.

Limitations

- qualitative rather than predictive model;
- no waning immunity;
- one vaccine type only;
- mutation operator is phenomenological;
- no age / space / asymptomatic structure.

Part II

Other approaches

RL Control (2D)

- Control both:
  - Vaccination rate n_t
  - NPI intensity η_t
- State = epidemic + variant distribution
- Goal: minimize infections, deaths, and evolution

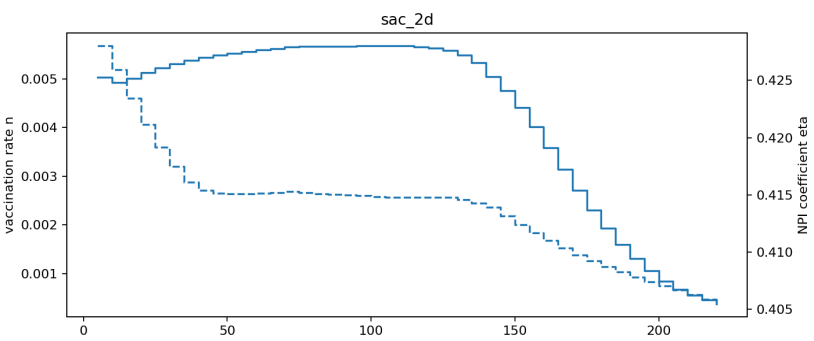


Figure 3

Algorithm (SAC)

- Continuous control (no discrete policies)
- Learns policy $\pi(a|s)$
- Balances:
  - epidemic suppression
  - evolutionary pressure

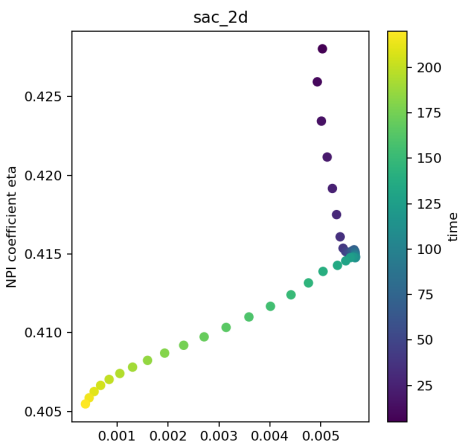
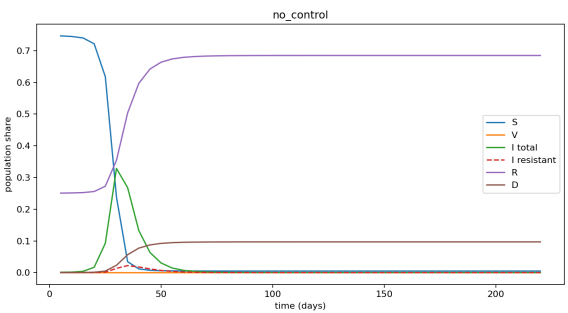
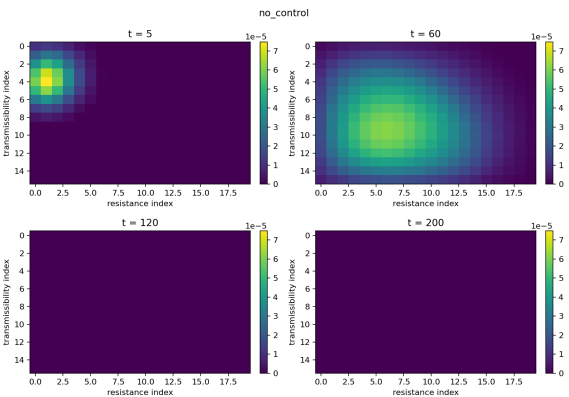


Figure 4

Baseline: No Control

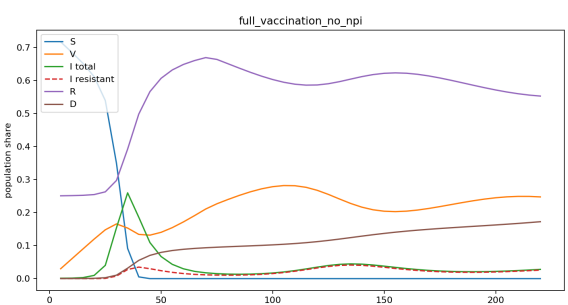
- Large outbreak
- No strong evolution

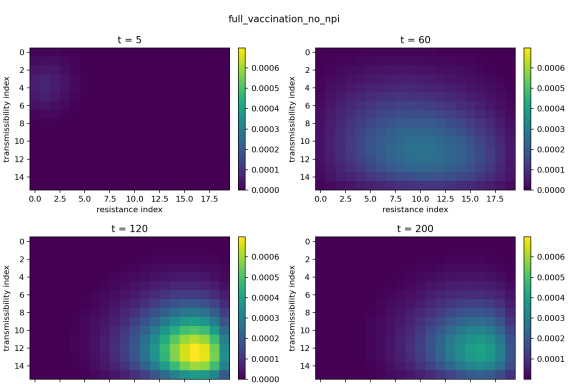




Baseline: Vaccination Only

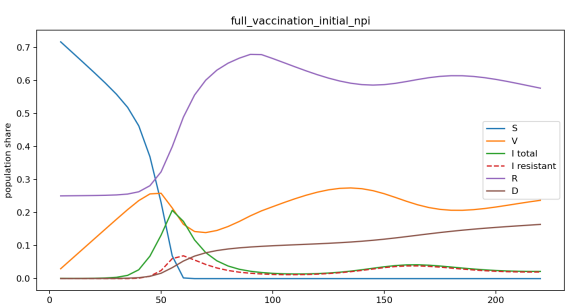
- Lower peak infection
- Higher deaths (!)
- Resistance emerges

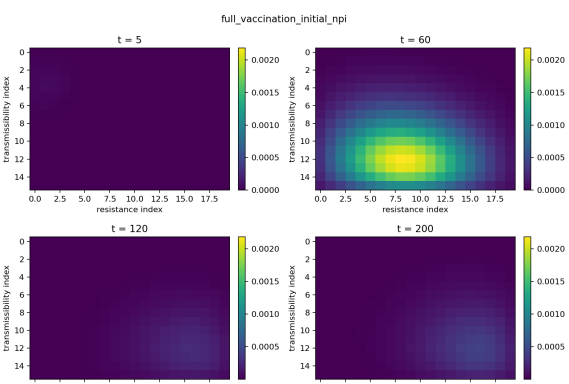




Baseline: Vaccination + NPI

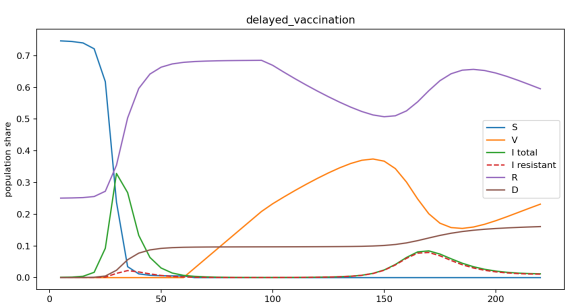
- Even lower infection
- Strong evolutionary pressure
- High resistance + transmissibility

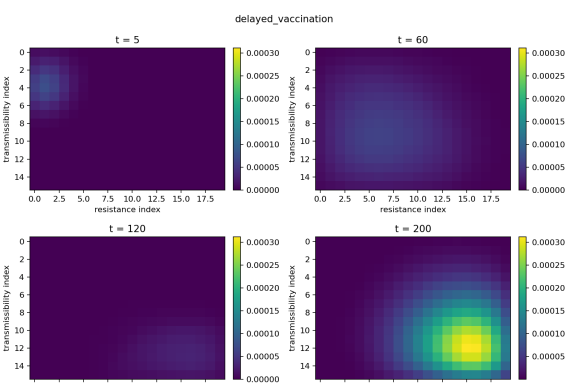




Delayed Vaccination

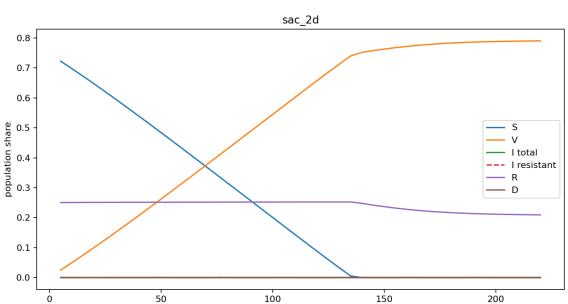
- Two waves
- Highest resistance

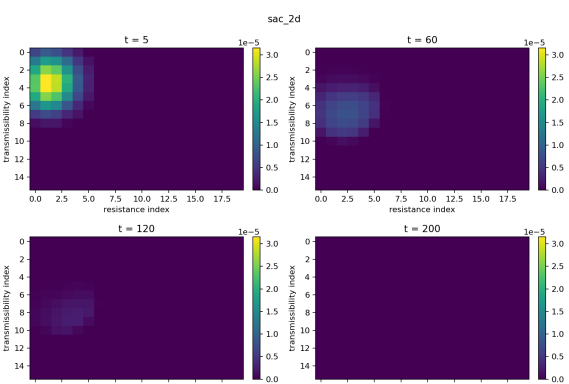




RL (SAC 2D) Results

- No large outbreak
- Near-zero deaths
- No resistant variants





Key Insight

- Strong static control $\Rightarrow$ strong evolution
- Weak control $\Rightarrow$ large outbreak
- RL $\Rightarrow$ balance both

Takeaway

Adaptive control prevents both epidemic spread and evolutionary escape